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Width of the circular ring (see *Fig. 4.21*)

= OB – OA= $r_2 - r_1$.

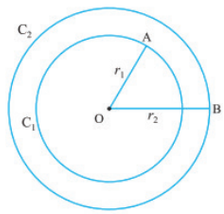


Fig. 4.21

4.3.2 Construction of Concentric Circles

Example Draw concentric circles with radii 4 cm and 6 cm and shade the circular ring. Find its width.

Step 1: Draw a rough diagram and mark the given measurements.

Step 2: Take any point O and mark it as the centre.

Step 3: With O as centre and draw a circle of radius OA = 4 cm

Step 4: With O as centre and draw a circle of radius OB = 6 cm.

Thus the concentric circles C_1 and C_2 are drawn.

Width of the circular ring = OB – OA = 6 – 4 = 2 cm.

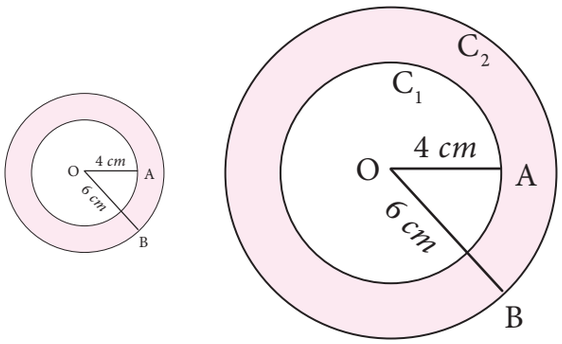


Fig. 4.22

Exercise 4.2

1. Draw circles for the following measurements of radius (r)/ diameters(d).

   (i) r = 4 cm (ii) d = 12 cm. (iii) r = 3.5 cm (iv) r = 6.5 cm.

   (v) d = 6 cm

2. Draw concentric circles for the following measurements of radii / diameters. Find out the width of each circular ring.

   (i) r = 3 cm and r = 5 cm.

   (ii) r = 3.5 cm and r = 6.5 cm.

   (iii) d = 6.4 cm and d = 11.6 cm.

   (iv) r = 5 cm and r = 7.5 cm.

   (v) d = 6.2 cm and r = 6.2 cm.

   (vi) r = 7.1 cm and d = 12 cm.

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